

FE CIVIL PRACTICE EXAM

16. A construction company bought a new rubber-tire loader and is performing a risk analysis about whether to purchase insurance. The construction company paid \$100,000 for the loader. The annual cost for the insurance premium is \$2,000, and the deductible is \$1,000. The risk options to purchase or not to purchase insurance are as follows:

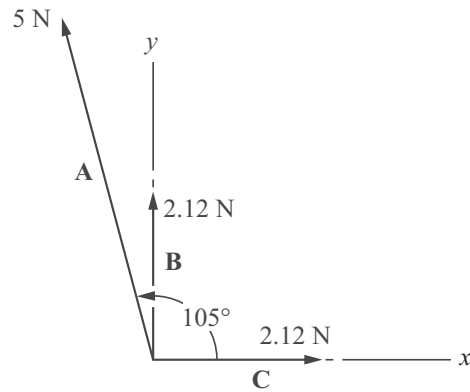
- 0.88 probability of no accident
- 0.11 probability of a small accident at a cost of \$800
- 0.01 probability of a total loss for the loader

The best option and projected cost savings are:

- ☐ A. purchase insurance and save \$990
 - ☐ B. purchase insurance and save \$1,088
 - ☐ C. do not purchase insurance and save \$1,010
 - ☐ D. do not purchase insurance and save \$2,098
17. A tractor cost \$7,500. After 10 years it has a salvage value of \$5,000. Maintenance costs are \$500 per year. If the interest rate is 10%, the equivalent uniform annual cost is most nearly:
- ☐ A. \$500
 - ☐ B. \$750
 - ☐ C. \$1,400
 - ☐ D. \$2,000

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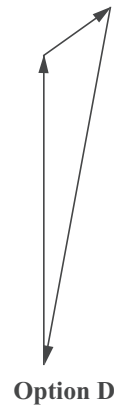
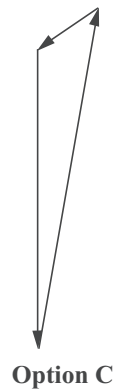
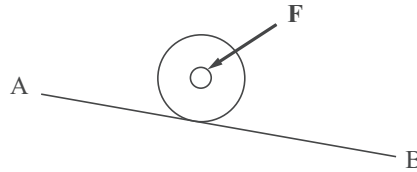
18. The magnitude (N) of the resultant of the three coplanar forces, A, B, and C, is most nearly:



- ☐ A. 7.0
- ☐ B. 7.8
- ☐ C. 9.2
- ☐ D. 10.3

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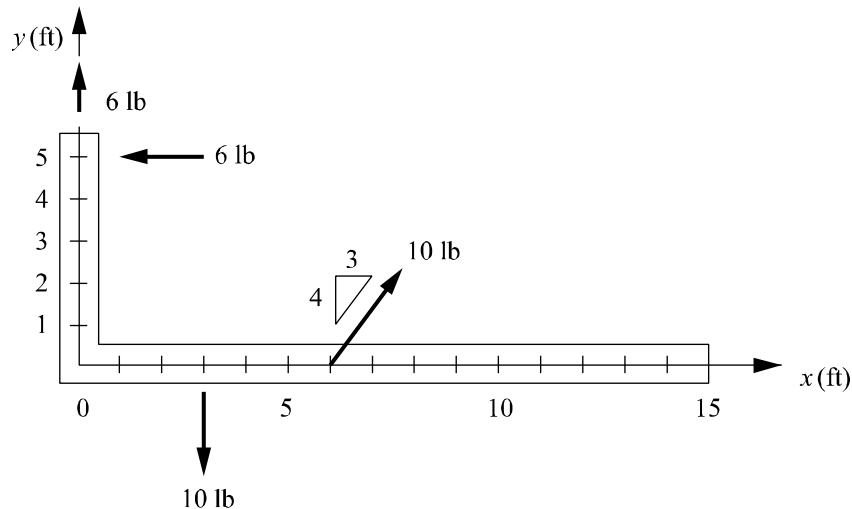
19. A heavy roller is held in equilibrium on a frictionless Plane AB by the force $\mathbf{F}$, as shown. Which diagram correctly shows a vector polygon of the forces acting on the roller?



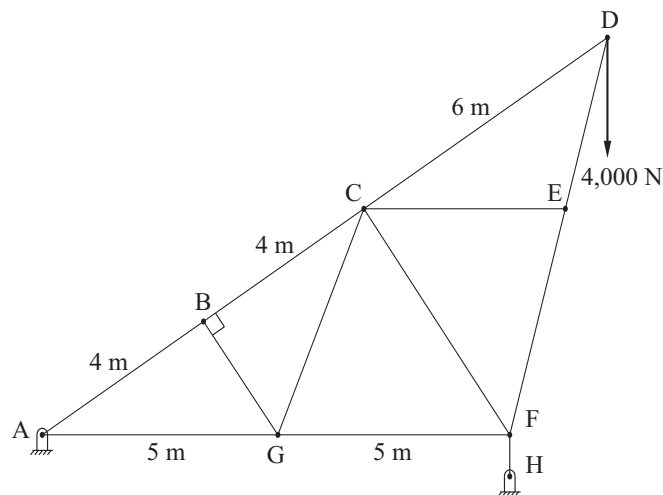
- ☐ A. Option A
- ☐ B. Option B
- ☐ C. Option C
- ☐ D. Option D

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20. Select the location on the x -axis where an additional load must be placed in order to achieve equilibrium of the L-shaped bar.



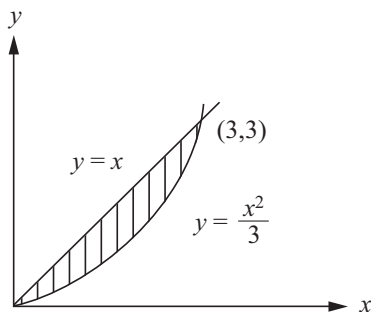
21. The figure below shows a simple truss. Which members in the truss have no (zero) force in them?



- ☐ A. BG, CG, CF, CE
- ☐ B. BG, CE
- ☐ C. CG, CF
- ☐ D. CF

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22. Consider the following graph:

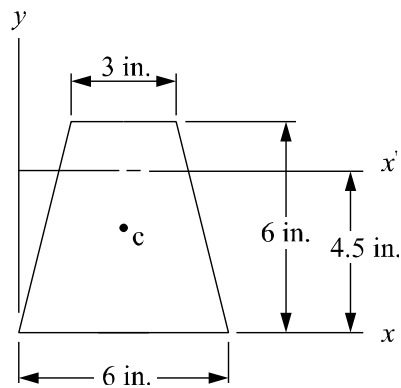


Which of the following expressions gives the distance from the y -axis to the centroid of the shaded area?

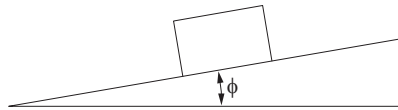
- A. $\frac{\int_0^3 \frac{1}{3} x^3 dx}{\int_0^3 \left(x + \frac{1}{3} x^2 \right) dx}$
- B. $\frac{\int_0^3 \left(x^2 - \frac{1}{3} x^3 \right) dx}{\int_0^3 \left(x - \frac{1}{3} x^2 \right) dx}$
- C. $\frac{\int_0^3 \left(x - \frac{1}{3} x^2 \right) dx}{\int_0^3 \left(x - \frac{1}{3} x^2 \right) dx}$
- D. $\frac{\int_0^3 \left(\frac{1}{2} x^2 + \frac{1}{3} x^3 \right) dx}{\int_0^3 \left(x - \frac{1}{3} x^2 \right) dx}$

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23. The moment of inertia (in^4) of the area about the x' axis ($I_{x'}$) is most nearly:



- ☐ A. 78.0
 - ☐ B. 78.6
 - ☐ C. 118.5
 - ☐ D. 138.8
24. In the figure below, the coefficient of static friction between the block and the inclined plane is 0.25. The block is in equilibrium. As the inclined plane is raised, the block will begin to slide when:

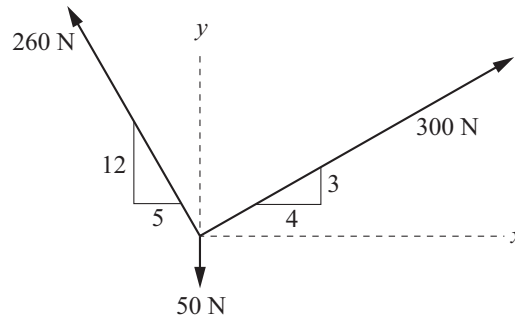


- ☐ A. $\sin \phi = 1.0$
- ☐ B. $\cos \phi = 1.0$
- ☐ C. $\cos \phi = 0.25$
- ☐ D. $\tan \phi = 0.25$

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25. Three forces act as shown below. The magnitude of the resultant of the three forces (N) is most nearly:

- ☐ A. 140
- ☐ B. 191
- ☐ C. 370
- ☐ D. 396



26. A boat accelerates at a constant rate of 12 ft/sec^2 . The boat travels 140 ft while its speed changes to 60 ft/sec. The initial velocity (ft/sec) was most nearly:

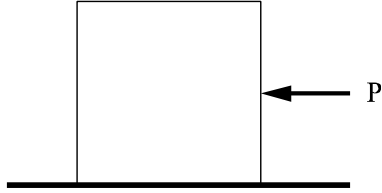
- ☐ A. 3.7
- ☐ B. 5.0
- ☐ C. 15.5
- ☐ D. 31.0

27. A small rotating robotic arm weighs 8 N and has a mass radius of gyration of 4.0 cm. The mass moment of inertia ($\text{kg}\cdot\text{cm}^2$) is most nearly:

- ☐ A. 13.0
- ☐ B. 32.0
- ☐ C. 128
- ☐ D. 256

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28. A 5-kg block is sliding along a frictionless surface and is acted on by a constant force **P** of 20 N. The time (seconds) when the block is moving at 18 m/s is most nearly:



- ☐ A. 0.22
 - ☐ B. 1.83
 - ☐ C. 4.50
 - ☐ D. 72.0
29. During impact of two objects, which of the following is true?
- ☐ A. Energy is never conserved.
 - ☐ B. Energy is always conserved.
 - ☐ C. Momentum is never conserved.
 - ☐ D. Momentum is always conserved.